

INDOCO REMEDIES LTD. Packaging Development Department

Item:	Leaflet
Item Code:	XXXXXXX
Open Size:	160 x 170
Folding Size:	H:Unfolded:160 mm Folded: 26.66 mm
	W:Unfolded:170 mm Folded:170 mm
GSM/ Paper:	54 gsm Maplitho Paper
Colour:	Black
Pharmacode:	
Layout No.:	CP 150 9919749-001

Reason for Preparation/ Revision:

1) Text updated

Prepared by	Checked by	Approved by

Path Y:\M\Metchek 850\Export\MALAYSIA\Goa 1\Current 23\Metchek 500 and 850 Leaflet XXXXXXX.cdr

R2

Date: 19.06.23 / 26.08.2025 / 09.09.2025

170 mm

	16 mm	10 mm	
	5mm		5mm
4mm			
26.66 mm			
16.002 mm			
4mm			

Pharma
code

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory

METCHEK 500/ 850
Metformin Tablets B.P.

COMPOSITION METCHEK TABLET 500 mg:

Each film coated tablet contains
Metformin Hydrochloride Ph. Eur 500 mg
Colour: Titanium dioxide

COMPOSITION METCHEK TABLET 850 mg:

Each film coated tablet contains
Metformin Hydrochloride Ph. Eur 850 mg
Colour: Titanium dioxide

DOSAGE FORM: Tablet

PRODUCT DESCRIPTION

METCHEK TABLET 500 mg and 850 mg:

METCHEK 500 mg: White to off white, round bevel-edged biconvex film coated tablets with '500mg' debossed on one side
METCHEK 850 mg: White to off white, round bevel-edged biconvex film coated tablets with '850mg' debossed on one side

PHARMACODYNAMICS :

Metformin is an antihyperglycemic agent, which improves glucose tolerance in type 2 diabetes subjects, lowering both basal and postprandial plasma glucose. Its pharmacologic mechanisms of action are different from those of sulfonylureas. Metformin decreases hepatic glucose production, decreases intestinal absorption of glucose and improves insulin sensitivity (increases peripheral glucose uptake and utilization). Unlike sulfonylureas, metformin does not produce hypoglycemia in either diabetic or nondiabetic subjects and does not cause hyperinsulinemia. With metformin therapy, insulin secretion remains unchanged while fasting insulin levels and daylong plasma insulin response may actually decrease. Monotherapy with metformin may be effective in patients who have not responded to sulfonylureas or who have only a partial response to sulfonylureas or who have ceased to respond to sulfonylureas. In such patients, if adequate glycemic control is not attained with Metformin monotherapy, the combination of metformin and a sulfonylurea may have a synergistic effect, since both agents act to improve glucose tolerance by different but complementary mechanisms. The magnitude of the decline in fasting blood glucose concentration following the institution of Metformin therapy is proportional to the level of fasting hyperglycemia. Non-insulin-dependent diabetics with higher fasting glucose concentrations will experience greater declines in plasma glucose and glycosylated hemoglobin. Metformin has a modest favorable effect on serum lipids, which are often abnormal in type 2 diabetes patients. In clinical studies, particularly when baseline levels were abnormally elevated, Metformin, alone or in combination with a sulfonylurea, lowered mean fasting serum triglycerides, total cholesterol and LDL cholesterol levels and had no adverse effects on other lipid levels. In contrast to sulfonylureas body weight of individuals on Metformin tends to remain stable or may be even decrease somewhat.

PHARMACOKINETICS :

Absorption: After an oral dose of metformin, t-max is reached in 2.5 hours. Absolute bioavailability of a 500mg or 850mg metformin tablet is approximately 50-60% in healthy subjects. After an oral dose, the non-absorbed fraction recovered in faeces was 20-30%. After oral administration, metformin absorption is saturable and incomplete. It is assumed that the pharmacokinetics of metformin absorption is non-linear.

At the usual metformin doses and dosing schedules, steady state plasma concentrations are reached within 24 to 48 hours and are generally less than 1 µg/ml. In controlled clinical trials, maximum metformin plasma levels (Cmax) did not exceed 4 µg/ml, even at maximum doses. Food decreases the extent and slightly delays the absorption of metformin. Following administration of a dose of 850 mg, a 40% lower plasma peak concentration, a 25% decrease in AUC (area under the curve) and a 35-minute prolongation of time to peak plasma concentration were observed. The clinical relevance of these decreases is unknown.

Distribution: Plasma protein binding is negligible. Metformin partitions into erythrocytes. The blood peak is lower than the plasma peak and appears at approximately the same time. The red blood cells most likely represent a secondary compartment of distribution. The mean Vd ranged between 63-276 L.

Metabolism: Metformin is excreted unchanged in the urine. No metabolites have been identified in humans.

Elimination: Renal clearance of metformin is > 400 ml/min, indicating that metformin is eliminated by glomerular filtration and tubular secretion. Following an oral dose, the apparent terminal elimination half-life is approximately 6.5 hours. When renal function is impaired, renal clearance is decreased in proportion to that of creatinine and thus the elimination half-life is prolonged, leading to increased levels of metformin in plasma.

Paediatrics:

Single dose study: After single doses of metformin 500 mg, paediatric patients have shown similar pharmacokinetic profile to that observed in healthy adults.

Multiple dose study: Data are restricted to one study. After repeated doses of 500 mg BID for 7 days in paediatric patients the peak plasma concentration (Cmax) and systemic exposure (AUC0-t) were reduced by approximately 33% and 40%, respectively compared to diabetic adults who received repeated doses of 500 mg BID for 14 days. As the dose is individually titrated based on glycaemic control, this is of limited clinical relevance.

INDICATIONS :

Treatment of type 2 diabetes mellitus, particularly in overweight patients, when dietary management and exercise alone does not result in adequate glycaemic control

In adults, Metckek film-coated tablets may be used as monotherapy or in combination with other oral anti-diabetic agents or with insulin.

In children from 10 years of age and adolescents, Metckek film-coated tablets may be used as monotherapy or in combination with insulin.

RECOMMENDED DOSE ADULTS :

Monotherapy and combination with other oral anti-diabetic agents:

The usual starting dose is one tablet 2 or 3 times daily given during or after meals.

After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastrointestinal tolerability. The maximum recommended dose of metformin is 3 g daily.

If transfer from another oral anti-diabetic agent is intended: discontinue the other agent and initiate metformin at the dose indicated above.

Combination with insulin:

Metformin and insulin may be used in combination therapy to achieve better blood glucose control. Metformin is given at the usual starting dose of one tablet 2-3 times daily, while insulin dosage is adjusted on the basis of blood glucose measurements.

Elderly: Due to the potential for decreased renal function in elderly subjects, the metformin dosage should be adjusted based on renal function. Regular assessment of renal function is necessary (see section 4.4).

Children and adolescents:

Monotherapy and combination with insulin

Metformin film coated tablets can be used in children from 10 years of age and adolescents.

The usual starting dose is one tablet of 500 mg or 850 mg once daily, given during meals or after meals.

After 10 to 15 days the dose should be adjusted on the basis of blood glucose measurements. A slow increase of dose may improve gastrointestinal tolerability. The maximum recommended dose of metformin is 2 g daily, taken as 2 or 3 divided doses.

Products containing Metformin as a single active ingredient:

Renal impairment

A GFR should be assessed before initiation of treatment with metformin containing products and at least annually thereafter. In patients at an increased risk of further progression of renal impairment and in the elderly, renal function should be assessed more frequently, e.g. every 3-6 months.

GFR mL/min	Total maximum daily dose (to be divided into 2-3 daily doses)	Additional considerations
60-89	3000 mg	Dose reduction may be considered in relation to declining renal function.
45-59	2000 mg	Factors that may increase the risk of lactic acidosis should be reviewed before considering initiation of metformin. The starting dose is at most half of the maximum dose.
30-44	1000 mg	
<30	-	Metformin is contraindicated.

ROUTE OF ADMINISTRATION :

Tablet for oral consumption

CONTRAINDICATION :

Hypersensitivity to metformin hydrochloride or to any of the excipients

Diabetic pre-coma

Severely reduced kidney function (GFR <30 mL/min)

Acute conditions with the potential to alter renal function such as -dehydration

170 mm

-severe infection
-shock
-Intra vascular administration of Iodinated contrast agents (see 4.4 Warnings and special precautions for sue).
Acute or chronic disease, which may cause tissue hypoxia such as:
Cardiac or respiratory failure
Recent myocardial infraction
Shock
Hepatic insufficiency, acute alcohol intoxication, alcoholism
Lactation
Any type of acute metabolic acidosis (such as lactic acidosis, diabetic ketoacidosis)

WARNING AND PRECAUTIONS :

Lactic acidosis : Lactic acidosis, a very rare but serious metabolic complication, most often occurs at acute worsening of renal function or cardiorespiratory illness or sepsis. Metformin accumulation occurs at acute worsening of renal function and increases the risk of lactic acidosis.
In case of dehydration (severe diarrhoea or vomiting, fever or reduced fluid intake), metformin should be temporarily discontinued and contact with a health care professional is recommended.
Medicinal products that can acutely impair renal function (such as antihypertensives, diuretics and NSAIDs) should be initiated with caution in metformin-treated patients. Other risk factors for lactic acidosis are excessive alcohol intake, hepatic insufficiency, inadequately controlled diabetes, ketosis, prolonged fasting and any conditions associated with hypoxia, as well as concomitant use of medicinal products that may cause lactic acidosis.

Patients and/or care-givers should be informed of the risk of lactic acidosis. Lactic acidosis is characterised by acidotic dyspnoea, abdominal pain, muscle cramps, asthenia and hypothermia followed by coma. In case of suspected symptoms, the patient should stop taking metformin and seek immediate medical attention. Diagnostic laboratory findings are decreased blood pH (< 7.35), increased plasma lactate levels (>5 mmol/L) and an increased anion gap and lactate/pyruvate ratio.

Metformin may reduce vitamin B12 serum levels. The risk of low vitamin B12 levels increases with increasing metformin dose, treatment duration, and/or in patients with risk factors known to cause vitamin B12 deficiency. In case of suspicion of vitamin B12 deficiency (such as anemia or neuropathy), vitamin B12 serum levels should be monitored. Periodic vitamin B12 monitoring could be necessary in patients with risk factors for vitamin B12 deficiency. Metformin therapy should be continued for as long as it is tolerated and not contraindicated and appropriate corrective treatment for vitamin B12 deficiency provided in line with current clinical guidelines.

Renal function: As metformin is excreted by the kidney, serum creatinine levels should be determined before initiating treatment and regularly thereafter:

at least annually in patients with normal renal function, at least two to four times a year in patients with serum creatinine levels at the upper limit of normal and in elderly subjects. Decreased renal function in elderly subjects is frequent and asymptomatic. Special caution should be exercised in situations where renal function may become impaired, for example when initiating antihypertensive therapy or diuretic therapy and when starting therapy with an NSAID.

Administration of iodinated contrast agent :

As the intravascular administration of iodinated contrast materials in radiologic studies can lead to renal failure, metformin should be discontinued prior to, or at the time of the test and not reinstituted until 48 hours afterwards, and only after renal function has been re-evaluated and found to be normal.

Surgery :

Metformin hydrochloride should be discontinued 48 hours before elective surgery with general anaesthesia and should not be usually resumed earlier than 48 hours afterwards.

Children and adolescents :

The diagnosis of type 2 diabetes mellitus should be confirmed before treatment with metformin is initiated. No effect of metformin on growth and puberty has been detected during controlled clinical studies of one-year duration but no long-term data on these specific points are available. Therefore, a careful follow-up of the effect of metformin on these parameters in metformin-treated children, especially pre-pubescent children, is recommended.

Children aged between 10 and 12 years:

Only 15 subjects aged between 10 and 12 years were included in the controlled clinical studies conducted in children and adolescents. Although metformin efficacy and safety in children below 12 did not differ from efficacy and safety in older children, particular caution is recommended when prescribing to children aged between 10 and 12 years.

Other precautions:

All patients should continue their diet with a regular distribution of carbohydrate intake during the day. Overweight patients should continue their energy-restricted diet. The usual laboratory tests for diabetes monitoring should be performed regularly.
Metformin alone never causes hypoglycaemia, although caution is advised when it is used in combination with insulin or sulphonylureas.

INTERACTIONS WITH OTHER MEDICAMENTS:

Concomitant use not recommended :

Alcohol

Increased risk of lactic acidosis in acute alcohol intoxication, particularly in case of:

Fasting or malnutrition
Hepatic insufficiency
Avoid consumption of alcohol and alcohol-containing medications

Iodinated contrast agents :

Intravascular administration of iodinated contrast agents may lead to renal failure, resulting in metformin accumulation and a risk of lactic acidosis. Metformin should be discontinued prior to, or at the time of the test and not reinstituted until 48 hours afterwards, and only after renal function has been re-evaluated and found to be normal.

Combinations requiring precautions for use :

Glucocorticoids (systemic and local routes), beta-2-agonists, and diuretics have intrinsic hyperglycaemic activity. Inform the patient and perform more frequent blood glucose monitoring, especially at the beginning of treatment. If necessary, adjust the dosage of the anti-diabetic drug during therapy with the other drug and upon its discontinuation. ACE-inhibitors may decrease the blood glucose levels. If necessary, adjust the dosage of the anti-diabetic drug during therapy with the other drug and upon its discontinuation.

PREGNANCY AND LACTATION:

To date, no relevant epidemiological data are available. Animal studies do not indicate harmful effects with respect to pregnancy, embryonal or foetal development, parturition or postnatal development (see also section 5.3). When the patient plans to become pregnant and during pregnancy, diabetes should not be treated with metformin but insulin should be used to maintain blood glucose levels as close to normal as possible in order to lower the risk of foetal malformations associated with abnormal blood glucose levels. Metformin is excreted into milk in lactating rats. Similar data are not available in humans and a decision should be made whether to discontinue nursing or to discontinue metformin, taking into account the importance of the compound to the mother.

SIDE EFFECTS:

The following undesirable effects may occur under treatment with metformin. Frequencies are defined as follows: very common > 1/10; common > 1/100, < 1/10; uncommon > 1/1,000, < 1/100; rare > 1/10,000, < 1/1,000; very rare < 1/10,000 and isolated reports.

Metabolism and nutrition disorders: Common : Vitamin B12 decrease/deficiency

Decrease of vitamin B12 absorption with decrease of serum levels during long-term use of metformin. Consideration of such etiology is recommended if a patient presents with megaloblastic anemia.

Very rare: Lactic acidosis

Nervous system disorders:

Common: Taste disturbance

Gastrointestinal disorders:

Very common: Gastrointestinal disorders such as nausea, vomiting, diarrhoea, abdominal pain and loss of appetite. These undesirable effects occur most frequently during initiation of therapy and resolve spontaneously in most cases. To prevent them, it is recommended that metformin be taken in 2 or 3 doses during or after meals. A slow increase of the dose may also improve gastrointestinal tolerability.

Hepatobiliary disorders:

Isolated reports: Liver function tests abnormalities or hepatitis resolving upon metformin discontinuation.

Skin and subcutaneous tissue disorders:

Very rare: Skin reactions such as erythema, pruritus, urticaria In published and post marketing data and in controlled clinical studies in a limited paediatric population aged 10-16 years treated during 1 year, adverse event reporting was similar in nature and severity to that reported in adults.

SYMPTOMS AND TREATMENT OF OVERDOSE :

Hypoglycaemia has not been seen with metformin doses of up to 85g, although lactic acidosis has occurred in such circumstances. High overdose or concomitant risks of metformin may lead to lactic acidosis. Lactic acidosis is a medical emergency and must be treated in hospital. The most effective method to remove lactate and metformin is haemodialysis.

STORAGE CONDITION :

Store below 30° C and protect from light.

SHELF LIFE :

36 months from the date of manufacture.

PRESENTATION :

Metckek 500 and 850 are available as a blister of 10 tablets. 10 or 50 such blisters are packed in a carton along with the packaging insert.

Reg. No.: Metckek 500: MAL08031359AZ

Metckek 850: MAL08031360AZ

Manufactured by:

INDOCO REMEDIES LTD.

L-14, Verna Industrial Area, Verna, Goa 403 722.

Regd. Office: 166, C.S.T. Road, Mumbai 400 098, INDIA.

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